

WEIYU LIU

weiyuliu.com
weiyul@stanford.edu

RESEARCH INTEREST

My research studies generalization in robotics, with an emphasis on structured representation of world knowledge for long-horizon, language-conditioned manipulation. My work leverages human language as a source of prior knowledge and grounds it through physical interactions to support reasoning over object properties, spatial relations, and task procedures. The goal of my research is to enable robots to robustly operate in open environments and respond naturally to human instructions such as “make the tea.”

EDUCATION

Georgia Institute of Technology 2017 – 2022
Ph.D. in Robotics
Advisor: Sonia Chernova

Georgia Institute of Technology 2012 – 2017
Bachelor of Science, Electrical Engineering with Distinction, Minor in Computer Science

ACADEMIC and INDUSTRY POSITIONS

Stanford Vision and Learning Lab, Stanford University 2023 – 2025
Postdoctoral Scholar with Jiajun Wu

Seattle Robotics Lab, NVIDIA, 2021, 2022
Robotics Research Intern with Dieter Fox, Tucker Hermans, Chris Paxton, and Animesh Garg

PUBLICATIONS

Journal Articles

- [4] Foundation Models in Robotics: Applications, Challenges, and the Future
Roya Firoozi, Johnathan Tucker, Stephen Tian, Anirudha Majumdar, Jiankai Sun, **Weiyu Liu**, Yuke Zhu, Shuran Song, Ashish Kapoor, Karol Hausman, Brian Ichter, Danny Driess, Jiajun Wu, Cewu Lu, Mac Schwager
International Journal of Robotics Research (IJRR), 2024
- [3] A Survey of Semantic Reasoning Frameworks for Robotic Systems
Weiyu Liu^{*}, Angel Daruna^{*}, Maithili Patel, Kartik Ramachandruni, Sonia Chernova
Robotics and Autonomous Systems (RAS), 2023
- [2] Latent Space Planning for Multi-Object Manipulation with Environment-Aware Relational Classifiers
Yixuan Huang, Nichols Crawford Taylor, Adam Conkey, **Weiyu Liu**, Tucker Hermans
IEEE Transactions on Robotics (T-RO), 2023
- [1] GraspGPT: Leveraging Semantic Knowledge from a Large Language Model for Task-Oriented Grasping
Chao Tang, Dehao Huang Wenqi Ge, **Weiyu Liu**, Hong Zhang
IEEE Robotics and Automation Letters (RA-L), 2023

Conference Papers

- [14] Learning Composable Skills by Discovering Spatial and Temporal Structure with Foundation Models
Neil Nie, Wenlong Huang, Jiayuan Mao, Li Fei-Fei, **Weiyu Liu**[†], Jiajun Wu[†]
IEEE International Conference on Robotics and Automation (ICRA), 2025

- [13] MoMaGen: Generating Demonstrations under Soft and Hard Constraints for Multi-Step Bimanual Mobile Manipulation
Chengshu Li*, Mengdi Xu*, Arpit Bahety*, Hang Yin*,
Yunfan Jiang, Huang Huang, Josiah Wong, Sujay Garlanka, Cem Gokmen,
Ruohan Zhang, **Weiyu Liu**, Jiajun Wu, Roberto Martín-Martín, Li Fei-Fei
International Conference on Learning Representations (ICLR), 2026
- [12] ENACT: Evaluating Embodied Cognition with World Modeling of Egocentric Interaction
Qineng Wang*, Wenlong Huang*, Yu Zhou, Hang Yin, Tianwei Bao, Jianwen Lyu, **Weiyu Liu**,
Ruohan Zhang, Jiajun Wu, Li Fei-Fei, Manling Li
International Conference on Learning Representations (ICLR), 2026
- [11] LayoutVLM: Differentiable Optimization of 3D Layout via Vision-Language Models
Fan-Yun Sun*, **Weiyu Liu***, Siyi Gu, Dylan Lim, Goutam Bhat, Federico Tombari, Manling Li,
Nick Haber, Jiajun Wu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
- [10] Learning Compositional Behaviors from Demonstration and Language
Weiyu Liu*, Neil Nie*, Ruohan Zhang, Jiayuan Mao[†], Jiajun Wu[†]
Conference on Robot Learning (CoRL), 2024
CoRL Workshop on Learning Effective Abstractions for Planning, 2024 (**Oral Presentation**)
- [9] Learning Planning Abstractions from Language
Weiyu Liu*, Geng Chen*, Joy Hsu, Jiayuan Mao, Jiajun Wu
International Conference on Learning Representations (ICLR), 2024
- [8] Naturally Supervised 3D Visual Grounding with Language-Regularized Concept Learners
Chun Feng, Joy Hsu, **Weiyu Liu**, Jiajun Wu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024
- [7] Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making
Manling Li, Shiyu Zhao, Qineng Wang, Kangrui Wang, Yu Zhou, Sanjana Srivastava, Cem Gokmen,
Tony Lee, Li Erran Li, Ruohan Zhang, **Weiyu Liu**, Percy Liang, Li Fei-Fei, Jiayuan Mao, Jiajun Wu
Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2024 (**Oral Presentation**)
Southern California Natural Language Processing Symposium, 2024 (**Best Paper**)
- [6] IKEA Manuals at Work: 4D Grounding of Assembly Instructions on Internet Videos
Yunong Liu, Shubh Khanna, Cristobal Eyzaguirre, Manling Li, Juan Carlos Niebles, Vineeth Ravi,
Saumitra Mishra, **Weiyu Liu**[†], Jiajun Wu[†]
Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2024
- [5] MARPLE: A Benchmark for Long-Horizon Inference
Emily Jin, Zhuoyi Huang, Jan-Philipp Fränken, **Weiyu Liu**, Hannah Cha, Erik Brockbank,
Sarah A Wu, Ruohan Zhang, Jiajun Wu, Tobias Gerstenberg
Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2024
- [4] Composable Part-Based Manipulation
Weiyu Liu, Jiayuan Mao, Joy Hsu, Tucker Hermans, Animesh Garg, Jiajun Wu
Conference on Robot Learning (CoRL), 2023
- [3] StructDiffusion: Language-Guided Creation of Physically-Valid Structures using Unseen Objects
Weiyu Liu, Yilun Du, Tucker Hermans, Sonia Chernova, Chris Paxton
Robotics: Science and Systems (RSS), 2023
- [2] Task-Oriented Grasp Prediction with Visual-Language Inputs
Chao Tang, Dehao Huang, Lingxiao Meng, **Weiyu Liu**, Hong Zhang
International Conference on Intelligent Robots and Systems (IROS), 2023
- [1] StructFormer: Learning Spatial Structure for Language-Guided Semantic Rearrangement of Novel Objects

Weiyu Liu, Chris Paxton, Tucker Hermans, Dieter Fox
International Conference on Robotics and Automation (ICRA), 2022

Workshop and Symposium Papers

- [3] Taking Recoveries to Task: Recovery-Driven Development for Recipe-based Robot Tasks
Siddhartha Banerjee*, Angel Daruna*, David Kent*, **Weiyu Liu***, Jonathan Balloch, Abhinav Jain,
Akshay Krishnan, Muhammad Asif Rana, Harish Ravichandar, Binit Shah, Nithin Shrivatsav, Sonia
Chernova
International Symposium on Robotics Research (ISRR), 2019.
- [2] SiRoK: Situated Robot Knowledge - Understanding the Balance Between Situated Knowledge
and Variability
Sonia Chernova, Vivian Chu, Angel Daruna, Haley Garrison, Meera Hahn, Priyanka Khante,
Weiyu Liu, Andrea Thomaz
AAAI Spring Symposium Series (AAAI-SSS), 2018.
- [1] Situated Bayesian Reasoning Framework for Robots Operating in Diverse Everyday Environments
Sonia Chernova, Vivian Chu, Angel Daruna, Haley Garrison, Meera Hahn, Priyanka Khante,
Weiyu Liu, Andrea Thomaz
International Symposium on Robotics Research (ISRR), 2017.

TEACHING

Georgia Institute of Technology	Fall 2021
<i>Graduate Teaching Assistant for Sonia Chernova and Sean Wilson</i>	
CS 7785 Intro Robotics Research	
Georgia Institute of Technology	Spring 2018
<i>Graduate Teaching Assistant for Sonia Chernova</i>	
CS 3630 Introduction to Robotics and Perception	
Georgia Institute of Technology	Fall 2014
<i>Peer Tutor</i> (Best Tutor Award)	
Learning Assistance Program	

LEADERSHIP and PROFESSIONAL SERVICE

Contributor to <i>A Roadmap for US Robotics</i> (national roadmap)	2024
Organizer for Workshop on Foundation Models for Embodied Agents at <i>CVPR</i>	2025
Organizer for Workshop on Vision-Language Models for Navigation and Manipulation at <i>ICRA</i>	2024
Georgia Institute of Technology, <i>Vice President Academic, RoboGrads</i>	2019 – 2020
Area Chair for <i>CoRL</i>	2025
Program Committee Member for RSS Pioneers	2023
Reviewer for <i>RSS</i>	2021, 2023, 2024
Reviewer for <i>CoRL</i>	2021 – 2024
Reviewer for <i>ICRA</i>	2019 – 2024
Reviewer for <i>RA-L</i>	2020 – 2024
Reviewer for <i>IROS</i>	2022 – 2024
Reviewer for <i>ICLR</i>	2024
Reviewer for <i>ECCV</i>	2024
Reviewer for <i>HRI</i>	2024
Reviewer for <i>EMNLP</i>	2023
Reviewer for <i>COLING</i>	2022
Reviewer for <i>AAMAS</i>	2022

AWARDS and HONORS

Best Paper, <i>SoCal NLP</i> (top 0.4%)	2024
RSS Pioneers	2022
Fetch it! The Mobile Manipulation Challenge, <i>First Place</i>	2019
Georgia Institute of Technology, <i>Dean's List</i>	2012 – 2017

TALKS

Invited Talk, Berkeley Artificial Intelligence Research Lab, UC Berkeley	2023
Invited Talk, Stanford Vision and Learning Lab, Stanford University	2022
Invited Talk, RoboGrads Student Seminar, Georgia Tech	2022
Invited Talk, Toronto AI in Robotics Seminar, University of Toronto	2022
Invited Talk, Laboratory for Progress, University of Michigan	2022
Oral Presentation, Workshop on Learning Effective Abstractions for Planning at <i>CoRL</i>	2024
Invited Talk, Robot Learning Workshop at <i>NeurIPS</i>	2021
Oral Presentation, Workshop on Semantic Representations at <i>ICRA</i>	2021
Oral Presentation, <i>AAAI</i>	2020
Oral Presentation, <i>ISRR</i>	2019

MENTORSHIP

Emily Jin (Stanford BS), PhD, Stanford	2023–Present
Yixuan Huang (University of Utah PhD), postdoc, Princeton	2022–2023
Chao Tang (Georgia Tech MS), postdoc, KTH	2022–2023
Xingjian Bai (Stanford Undergraduate Summer Exchange), PhD, MIT	2023
Chun Feng (Stanford Undergraduate Summer Exchange), MS, UIUC	2023
Chen Geng (Stanford Visiting Student), MS, UCSD	2023
Dhruva Bansal (Georgia Tech undergraduate), MS, Stanford	2021
Neil Nie (Stanford MS), Co-Founder & CEO, Verne Robotics	2023–2025
Fan-Yun Sun (Stanford PhD), Co-Founder & CEO, Moonlake AI	2024–2025
Zhuoyi Huang (Stanford MS), Member of Technical Staff, xAI	2023–2024
Yunong Liu (Stanford MS), Research Engineer, Luma AI	2023–2024
Tarun Chiruvolu (Stanford MS), Engineer, Tesla	2024
Jayanta Bhowmick (Georgia Tech undergraduate), Software Engineer, Amazon	2021